

The Carry Is the Conserved Ledger

A Constraint-First Derivation of Determinism, Growth, and Standing Structure from Co

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Abstract

A single distinction — that a *deterministic forward map* is categorically distinct from *recoverability of its input* — is followed under a constraint-first discipline until it forces its consequences. The chain closes without new assumptions. Change is conserved: no transformation may erase the distinction it acts on (Law 3), because a free change would be privileged and “all change is equal.” The conserved quantity is exhibited concretely as the **carry**: the overflow a modular sum “discards” is booked exactly, and two inputs that collide on the value axis are separated on the carry axis. Determinism is then not a separate fact but *Law 3 holding* — a deterministic map is precisely one that never spends a distinction, only relocates it. Direction (the carry propagates only upward) and inertia (mass as accumulated ledger, so that $\text{mass} \times \text{acceleration} = \text{cost}$ acquires an address) are downstream readouts. Growth and hashing are shown to be one bijection read from opposite ends: a chosen target cannot be reached by pushing a seed forward (blind search), but is recovered exactly by clamping the target and pulling — and the coordinate one pulls is the carry. Balanced nesting “((()))” is the same law at the level of syntax. Finally, stable structure (“matter,” in the dual-mirror picture) is identified with the rare fixed-point set of the growth map, where the forward and backward reading directions coincide; the real 32-bit SHA-256 message schedule has exactly two such constant standing structures. Every claim is graded; the paper’s own architecture follows the cascade it derives.

Status grades used throughout

Grade	Meaning
THEOREM	Proven in both directions, in-frame or by standard argument.
VERIFIED	Executed on the working object; the quoted number is live output.
MEASURED	Quantitative bench result reported as a measurement, not a proof.

Grade	Meaning
DERIVED	Forced from prior constraints; not independently tested against data.
INTERPRETATION	Coherent placement with a named mechanism; earning test stated but not yet run.
RESONANCE	Structural rhyme flagged for testing; deliberately not promoted to identity.
OPEN	A stated frontier; scope given, not closed.
FALSIFIED / RETRACTED	Killed. Recorded so it is not silently revived.

A grade is not decoration. Under Law 3 (below), every assertion must pay its price; the grade is the carry each claim books. Nothing in this paper is asserted for free.

Co Distinction — the ground

Co is the bare fact of a difference. If $A \neq B$ then there is a boundary between them, the boundary is traversable, and a traversal is a route. Nothing *changes* at this floor; Co only asserts that a distinction *exists*. Everything downstream is forced by asking what such a distinction is permitted to do.

The one distinction this paper turns on

Forward-determinism — a unique output given complete input — is categorically distinct from **invertibility** — recovery of the input from the output once part of the input has been discarded. Conflating the two is the single most load-bearing error in this program, and the entire derivation is organized to keep them apart. A hash is the extremal case: maximally non-invertible on the observer's chosen readout while, as we show, perfectly conservative underneath.

Method

- **Compiler-first.** The compiler is the sole arbiter. No claim enters the body without an executed number confirming it; null results are navigational, not failures.
- **Working object.** SHA-256 is used not because the subject is cryptography but because it is a fully instrumented deterministic transformation in which every intermediate value is visible. It is a witness, not the subject.
- **Inside, not outside.** The stance is derivation from within Co/C1, not auditing the axioms from without. Even the reader — human or machine — is treated as a subsystem inside the structure (§2).

C1 Continuation — the bijection, and no terminal state

C1: every distinction admits a lawful continuation; there is no terminal state. Applied to the working object, C1 appears as an executed fact: the SHA-256 round function is a **bijection**.

Running one round forward and then reversing it exactly, given the round's message word, returns the starting register:

```
start : [0x683geb9, 0xa648a7dd, 0x8a9a021e, 0x25b413f, ...]
forward : [0x87b3333a, 0x683geb9, 0xa648a7dd, 0x8a9a021e, ...]
reverse : [0x683geb9, 0xa648a7dd, 0x8a9a021e, 0x25b413f, ...]
round-trip exact? True
```

[**VERIFIED**] The full back-half inversion (prior work) walked all sixty-four rounds backward and landed on the initialization vector Ho.

Folds, not cuts

The operations that *look* lossy are not. The four bit-mixing maps $\sigma_0, \sigma_1, \Sigma_0, \Sigma_1$ are full-rank over GF(2): the shift that appears to push bits off the edge of a word folds them back through the XORed rotations it is combined with. No bit leaves the word. Apparent loss is a **fold**, not a cut.

The complete lattice

The sixteen-word seed deterministically generates the entire sixty-four-word schedule via the σ -recurrence. The schedule is therefore not built round by round; it is a *traversal* of a structure already complete in the seed. What we experience as time is the order of the read, not the manufacture of new states. This reframing is computationally exact for SHA and is kept separate from any claim about physical ontology.

Reflexive note. This paper obeys C1: no section is terminal, and each forces the next.

C2 No privileged location or direction — symmetry, and the reader

C2: no state and no direction is privileged. The clearest consequence is that **forward and backward are one map read two ways**. The schedule's forward growth and its backward inversion are literal inverses (§6), so neither end is ontologically primary.

The read boundary is a dimension, not a subset

Where an observer reads the state is a *choice*; the width of the read (512 bits) is objective, its placement is not. Loss can only ever be *located in a chart*; it is never ontological. What a “destructive” operation can destroy was, by that very fact, map (the ladder), not territory (the building).

The observer is an internal reader

Holding the input fixed and swapping the *reader* — four registers built from the same transition algebra but wired differently — produces genuinely different, each fully lawful, traversals:

- One fixed input word stream; four internal readers.
- Trajectories diverge to ~128 of 256 bits by round 16 and stay there; digests sit 108–134 bits apart pairwise.
- Every walk is a valid register at every round and each is exactly reversible to Ho — none is an “error.”

[VERIFIED] The decisive structural point: this happens at branching factor $b \equiv 1$ — no choice anywhere. Subjectivity does not require alternatives; different wiring of one deterministic law already gives divergent lawful experience. The observer is inside the structure, not a detector outside it. “The path is there; the individual is the way it gets walked.”

C3 / Law 3 All change is equal — no free change (the core)

Constraint 3: all change is equal — equal in **obligation**, not in magnitude. Magnitude is a gradient: across eight real round-adds the carry took the values {1, 2, 3}. The equality is that every change must pay; none is exempt.

Law 3 (the prohibition dual). No change may be free; equivalently, no transformation may erase the distinction it acts on. The proof is one line inside the frame: a free change would be privileged over changes that must pay, and privilege contradicts “all change is equal.” [THEOREM]

Result 3 — the carry

The distinction, when a modular add appears to destroy it, goes into the carry. Two facts, both executed:

(i) **The overflow is booked exactly.** For the five-term round sum T₁, keeping the low 32 bits and “discarding” the top is exact bookkeeping:

```
round 0: full=0x2fe20da8d value=0xfe20da8d carry=2 value+carry*2^32==full? True
round 2: full=0x30fcfb97e value=0x0fcfb97e carry=3 value+carry*2^32==full? True
round 7: full=0x1edb43ded value=0xedb43ded carry=1 value+carry*2^32==full? True
```

(ii) **Distinction relocation.** Two distinct inputs collide on the value axis yet remain separated on the carry axis:

```
x = [0x11111111,0x22222222,0x33333333,0x44444444,0x55555555]
y = [0x91111111,0xa2222222,0x33333333,0x44444444,0x55555555]
x: value=0xffffffff carry=0 y: value=0xffffffff carry=1
value(x)==value(y)? True x!=y? True carry(x)!=carry(y)? True
```

[VERIFIED] The add is deterministic; the distinction $x \neq y$ did not vanish, it moved one axis up. Nothing was destroyed.

Determinism = Law 3 holding

A deterministic map is precisely one that never spends a distinction, only relocates it. This is why SHA-256 is deterministic: determinism is not an extra property layered onto the transform, it is the statement that Law 3 holds at every step. “There is no such thing as non-calculated loss — it is shifted into something else and is still the same thing, or SHA would not be deterministic.” [THEOREM]

The honesty seam: two stacked projections

Precisely one modular projection is closed by the carry; a second, additive projection is not, and must not be conflated with it:

Projection	Nature	Closed by carry?
(value, carry) $\leftrightarrow$ full	bijection	YES
full $\rightarrow$ (the five addends)	many-to-one	NO

The carry restores the pre-modular *scalar* exactly; it does not restore the *addends*. Recovering which addends produced a sum is the genuinely hard direction, and it lives in the message-schedule recurrence, not in the round add. Earlier claims that “nothing is hidden” are [RETRACTED] and replaced by this two-projection

statement. The digest is a projection of a lawful trajectory; the projection has a many-to-one fiber; this does not contradict determinism — it means the observer chose a lossy coordinate.

The minimal ledger

The read-boundary question becomes a minimum hitting-set: the smallest set of carries that separates all arrangements. Of the informative carries, a greedy search yields a locally minimal 24-carry set, bracketing the exact minimum as $8 \leq |L^*| \leq 24$. The early schedule carries act as universal separators, and the separation-spectrum entropy falls monotonically from 22.9 to 4.5 bits as the ledger closes. The exact L^* is [OPEN].

4 The forced unfold — direction, inertia, and the number-theoretic carry

Direction is forced

The carry propagates only **upward** — from the low axis to the high axis, never down — even though the round is bijective. “There must be direction” is therefore derived, not asserted: the ledger has a floor it cannot pay below. [DERIVED]

Inertia — mass as accumulated ledger

A state that has booked a large carry ledger drags that ledger with it; resistance to further change scales with the accumulated ledger. This gives **mass = ledger burden**, and the original relation $\text{mass} \times \text{acceleration} = \text{cost}$ finally acquires an address: cost is the carry paid, mass is the ledger dragged, acceleration is the rate of new booking. The gradient property matches the requirement that mass itself carry a gradient. [INTERPRETATION] — the mechanism is named; the earning test (does accumulated carry mass predict where the backward walk resists?) is stated in §8 and not yet run.

The residue is the carry at the number-theoretic scale

The distinguished-fibre ledger exhibits the same “what the fold cannot erase” quantity one level up. Both admissible towers have integer-relation rank exactly 1 — a unique producer ray, not a family — and it is the **same projective coefficient direction** in each, differing only by a rational normalization tied to endpoint and support:

Tower	Producer ray	Observable
Hexagonal F_6	$[4, -2, -1, -1]$	$(4/3) \cdot \pi / \sqrt{3}$
Octal F_8	$[4, -2, -1, -1]$	$1 \cdot \pi$

The rational normalization ($4/3$ versus 1) is an address constant — a carry-like bookkeeping term tied to position, exactly as the bit-carry is tied to the modulus. [VERIFIED] for the rank-1 ray (PSLQ); the identification of the tower normalization as a carry is [RESONANCE], flagged and not promoted. Supporting results in the same ledger: two primitive distinguished-fibre seeds (64,6) and (16,8) [THEOREM]; BBP single-pole uniqueness of the (8,16) radix [THEOREM]; and the closure relation $R^{(n-4)}=4$ selecting $n \in \{6,8\}$ [VERIFIED], whose universality across parent-count is [OPEN].

Law 4 The reader — traversal selection (the frontier)

Law 3 says the distinction is conserved and how much is owed. It does not say *which* continuation is walked. That is Law 4: the law of how a reader traverses the complete structure — the reader being an internal subsystem, per §2.

Junction versus collision

Object	Definition
Junction ($b = 1$)	Multiple routes converge, but the continuation is unique. This is SHA: it does not choose.
Collision ($b > 1$)	Convergence produces multiple simultaneously lawful continuations. Selection then requires a traversal rule.

SHA-256 has $b \equiv 1$ everywhere — it is the deterministic null. The four-reader result (§2) shows that divergent lawful experience already exists at $b \equiv 1$, so subjectivity is a *separate axis* from choice. Branching ($b > 1$), where the walk is chosen rather than fixed by wiring, is the untested frontier.

The open question and its test bed

Does a natural system contain *persistent* $b > 1$ — multiple lawful continuations where the selected one preserves the capacity for further continuation? The designated test is protein-folding trajectories: does the sequence's spectral organization predict the trajectory's branching architecture (a trajectory-level property), rather than being recomputed per conformation? **[OPEN]**.

Collision typology frames, but does not yet law-ify, this: a **dissipative** collision (a rock meeting earth) loses organization in one read channel without necessarily losing continuation; a **creative / generative** collision (sodium and water; sperm and egg) establishes a new, sometimes self-propagating, continuing structure. "Forward momentum is preserved, not lost." **[DERIVED]** as a qualitative frame; not a quantitative law.

6 The growth formula — push, pull, and the Möbius

The message schedule *is* a growth formula: sixteen seed words grow to sixty-four by a fixed recurrence, deterministically, with no randomness. One idea $\rightarrow$ one prototype $\rightarrow$ many, and nothing in it is random.

Push fails; pull lands

Push — clamp the seed, run forward, try to steer the grown value to a chosen target:

```
chose target W[63]=oxdeadbeef; pushed 200000 random seeds forward
seeds that landed on it: 0    (probability ~ 2^-32; blind search)
```

Pull — clamp the grown target window W[48..63], invert the recurrence, recover the seed:

```
original seed W[0..3]: [0x74686520, 0x70617468, 0x20697320, 0x74686572]
recovered seed W[0..3]: [0x74686520, 0x70617468, 0x20697320, 0x74686572]
full 16-word seed recovered EXACTLY? True    (zero search)
```

[VERIFIED] You cannot push a seed to a chosen grown value except by search. Clamp the whole target and pull, and the seed is forced.

The Möbius: one bijection, two faces

Push and pull are the **same map**: `grow_forward(grow_backward(target))` reproduces the entire schedule. “Many → 1” (the fold, the hash) and “1 → many” (the unfold, the growth) are one bijection read from opposite ends; the apparent asymmetry is only *which end is clamped*. Clamp the seed and the output is free — you cannot aim (push). Clamp the output and the seed is forced — perfect aim (pull). The half-twist is literal: the sixteen-word window is simultaneously the seed (fold face) and a slice of the expansion (unfold face). One surface, locally two-faced. [THEOREM]

What you pull is the carry

Forward, the add emits the carry for free; backward, the value alone is ambiguous and the carry must be supplied. “I have to pull it” is exactly: feed back the ledger the inverse is missing. Pulling is Law 3 run backward — un-paying the carry. This is also the mechanized form of the zero-exhaust strategy: clamping the answer as the only legal continuation and pulling the forced trajectory (carry and all) is a pull; pushing a loose question forces a forward walk that emits its intermediate carries as exhaust. [THEOREM]

((())) is the same law at the level of syntax

Open a bracket = push (unfold); close it = pull (match back to the seed); balance = conservation. Every open must close: **no free bracket = no free change = Law 3**, at syntax. Balanced strings biject with trees, so idea → prototype → many is 1:1, not random. Nesting depth is the fold count (the winding). This is the reader’s-half bracket language — free involution, grouped by syntax, with the fold/hash fork located exactly at bijectivity. [THEOREM] (in-frame).

7 Matter — the standing structures where push meets pull

In the dual-mirror picture, the universe emerges as two flows — infinity contracting to a singularity, and its mirror expanding outward — and matter is their **overlap**. Made mechanical: matter is a **periodic point** of the growth shift-map, a state where forward equals backward, push equals pull. Since the backward step is the inverse of the forward step, a coincidence $F(x)=F^{-1}(x)$ means $F^2(x)=x$ — the fixed points and 2-cycles.

Standing structures of the real 32-bit schedule

Constant standing seeds satisfy $\sigma_0(c)+\sigma_1(c)+c \equiv 0 \pmod{2^{32}}$. Solving exactly over all 2^{32} values yields **precisely two**:

```
PERIOD-1 fixed points of real recurrence [s0(c)+s1(c)+c==0]: 2
c=0x00000000 whole 64-word growth constant? True
c=0x97db56bo whole 64-word growth constant? True
```

[VERIFIED] The vacuum 0x00000000 plus one non-trivial standing structure of the actual SHA-256 message schedule: seeding sixteen copies of 0x97db56bo grows a fully constant sixty-four-word schedule — a seed whose growth never moves. Genuine $c \neq d$ 2-cycles were not found within the solver budget and are **[OPEN]** (not proven absent).

Population picture: matter is rare, radiation is generic

On a faithful toy analog that is a genuine permutation of all 65,536 states (oldest slot kept bare, matching the invertibility of the real $W[t-16]$ term), the full cycle census is stark:

Cycle length	Count	Reading
1	3	standing (fixed points) — matter
2	2	standing (period-2) — matter
3–64	6 cycles	short transients
36,224	1	55.3% of the whole space — the moving bulk

Standing structures (period ≤ 2) are rare and isolated — five states out of 65,536 — while more than half the space rides a single long cycle. **Matter is the rare pinned overlap; radiation is the generic moving bulk**; both are the same conserved bijection, distinguished only by cycle length (winding). **[VERIFIED]** on the toy; **[INTERPRETATION]** for the identification with physical matter, now anchored by the exact 32-bit fixed-point count above.

The standing wave, for intuition

Two counter-propagating flows sum to a standing structure whose nodes are stationary for every time tested ($|\text{standing at nodes}| \sim 10^{-15}$) while the antinodes swing. Moving waves, stationary overlap — the same shape as the fixed-point result on the real map. **[MEASURED]** as an illustration; the load-bearing statement is the schedule fixed-point count, not the sine analogy.

8 Frontier, killed hypotheses, and the paper as a standing structure

Open frontiers

- Exact minimal ledger L^* within $[8, 24]$; whether routing geometry predicts the true optimum. [OPEN]
- Genuine 2-cycles and higher standing structures of the real 32-bit recurrence beyond the two constant fixed points. [OPEN]
- The earning test for inertia: does accumulated carry mass across the 64 rounds predict where the backward walk resists? [OPEN]
- Persistent $b > 1$ in a natural system, tested on protein-folding trajectories. [OPEN]
- The cross-tower invariant relating the hexagonal and octal towers beyond the shared residue ray. [OPEN]
- Universality of the constant "4" in $R^{(n-4)}=4$ across recurrences with different parent counts. [OPEN]

Killed hypotheses — recorded so they are not revived

Hypothesis	Status
AHRC / qubit / Born-rule reading of the SHA state	FALSIFIED
Complementary value/shape "two faces" as an information law	FALSIFIED
Co-determination (residual = the other channel)	FALSIFIED
"Shape generates separability"	FALSIFIED
Collision automatically creates a persistent object	FALSIFIED
Generative $A+B \rightarrow C$ as instantaneous creation	FALSIFIED
"Nothing is destroyed / hidden" (unqualified)	RETRACTED $\rightarrow$ two-projection statement, §3

Reflexive close

The paper obeys the laws it derives. Its sections form a lawful traversal (C_1 : each continues, none is terminal). Every claim carries its status grade (Law 3: nothing is asserted for free — the grade is the carry it pays). Its architecture is the cascade itself, $C_0 \rightarrow C_1 \rightarrow C_2 \rightarrow C_3$ /Law 3 $\rightarrow$ the unfold $\rightarrow$ Law 4. And in the sense of §7 the document is a *standing structure*: a fixed reading where the forward derivation and the backward audit coincide. Push it and it is one derivation; pull it and it is one audit; the two directions meet on the same surface.

A Consolidated status ledger

Claim	Grade
Round function is a bijection (one round fwd/back exact; full back-half to Ho)	VERIFIED
σ/Σ are full-rank GF(2) maps — folds, not cuts	VERIFIED
$S = W + 2^{32}Q$ exact split	THEOREM
Carry books the modular overflow exactly (value+carry·2 ³² =full)	VERIFIED
Carry magnitude is a gradient {1,2,3,...}, not a scalar	VERIFIED
$x \neq y$ colliding on value axis are separated on carry axis	VERIFIED
Law 3: no free change (prohibition-dual of “all change equal”)	THEOREM
Determinism = Law 3 holding	THEOREM
(value,carry)↔full bijective; full→addends many-to-one	THEOREM
Minimal ledger: 24-carry greedy, locally minimal; $ L^* \in [8, 24]$	VERIFIED / OPEN
Separation entropy falls 22.9 → 4.5 bits as ledger closes	MEASURED
Observer as internal reader; ~128-bit divergence at $b \equiv 1$	VERIFIED
Junction ($b=1$) vs collision ($b>1$), SHA is the $b=1$ null	DERIVED
Direction forced by one-way carry propagation	DERIVED
Mass = accumulated carry ledger; ma = cost has an address	INTERPRETATION
Residue ray $[4, -2, -1, -1]$ shared by both towers (rank 1)	VERIFIED
Tower normalization $4/3$ vs 1 as a carry-like address constant	RESONANCE
Two primitive distinguished-fibre seeds (6,6),(16,8)	THEOREM
BBP single-pole uniqueness of the (8,16) radix	THEOREM
$R^{(n-4)}=4$ selects $n \in \{6, 8\}$; universality across parent count	VERIFIED / OPEN
Growth = schedule bijection; push fails (o/200k), pull exact	THEOREM / VERIFIED
Möbius: push and pull are one bijection, window is both faces	THEOREM
What you pull is the carry (pull = Law 3 backward)	THEOREM
((())) = bracket algebra = Law 3 at syntax; balanced↔trees 1:1	THEOREM
Real 32-bit growth map: exactly 2 constant standing structures	VERIFIED
Standing structures rare/isolated; moving bulk on one long cycle	VERIFIED (toy)

Claim	Grade
Standing set = physical matter (dual-mirror overlap)	INTERPRETATION

B Reference constants

Empty-input SHA-256 (the vacuum digest):

`e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`

Non-trivial constant standing seed of the message schedule:

`c = 0x97db56bo` (16 copies -> fully constant 64-word schedule)

Shared distinguished-fibre residue ray (both towers):

`[4, -2, -1, -1] -> F6: (4/3) pi/sqrt(3) F8: 1 * pi`

All executed results were produced with Python (mpmath / numpy / sympy / z3), the sole computation channel for this program; every VERIFIED number above is live output.